Practice workbook · Describe how a named infection spreads, shows itself and is prevented

Transmission routes, reservoirs, incubation, symptoms and detection, and prevention measures for Ebola, HIV, cholera, gonorrhoea, pinworm and diarrhoea, including blood-borne risks.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DISEASE-TRANSMISSION-PREVENTION

1. Question 3 · part b · 3 marks

June 2019 Paper 1R

Worked-example retry · Attempt part b only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

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(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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1. Question 3 · part b · 3 marks · Working and answer

Working and answer

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2. Question 7 · part b · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b only.

7 Cholera is a disease that causes chronic water loss from the body.

(a) Name the type of microorganism that causes cholera.

(1)

(b) Describe how cholera is transmitted from one person to another.

(2)

(c) Explain how a person with cholera can be treated for the disease.

(4)

(Total for Question 7 = 7 marks)

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2. Question 7 · part b · 2 marks · Working and answer

Working and answer

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3. Question 8 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

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8 (a) Cholera is a disease that kills approximately 95 000 people each year worldwide.

(i) Name the type of pathogen that causes cholera.

(1)

(ii) State how cholera is transmitted from one person to another.

(1)

(b) Describe a method used to treat a person with cholera.

(3)

(Total for Question 8 = 5 marks)

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Working and answer

4. Question 1 · part b · 4 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

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(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

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The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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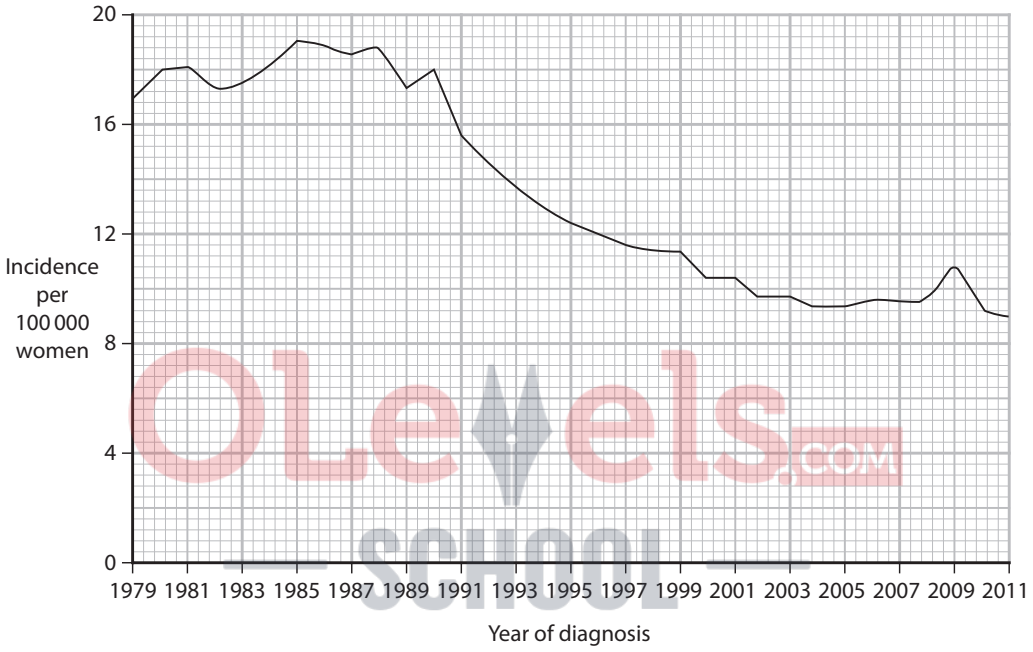
4. Question 1 · part b · 4 marks

June 2024 Paper 2

Further practice · Attempt part b only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.



(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

(2)

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4. Question 1 · part b · 4 marks · Working and answer

Working and answer

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5. Question 4 · part a · 2 marks

November 2024 Paper 2

Further practice · Attempt part a only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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5. Question 4 · part a · 2 marks

November 2024 Paper 2

Further practice · Attempt part a only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

5. Question 4 · part a · 2 marks · Working and answer

Working and answer

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6. Question 4 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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6. Question 4 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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7. Question 4 · part e · 2 marks

January 2020 Paper 2

Further practice · Attempt part e only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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7. Question 4 · part e · 2 marks

January 2020 Paper 2

Further practice · Attempt part e only.

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(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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7. Question 4 · part e · 2 marks

January 2020 Paper 2

Further practice · Attempt part e only.

(e) Suggest how an infected person and other people in the house can help prevent the spread of a pinworm infection (lines 13 to 14).

(2)

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(Total for Question 4 = 12 marks)

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Working and answer

8. Question 2 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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8. Question 2 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

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8. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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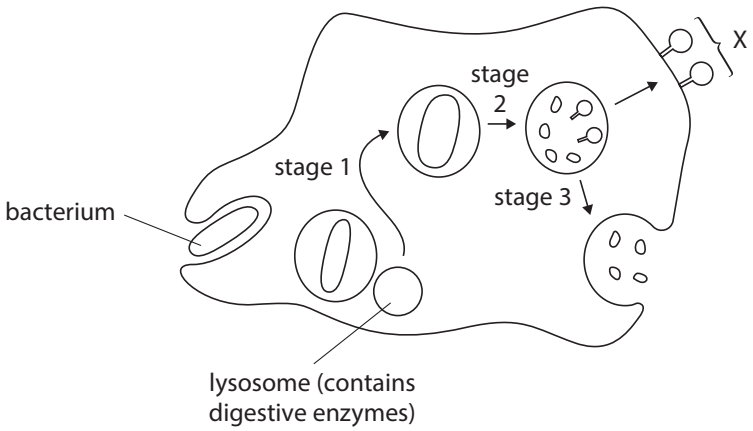
9. Question 6 · part b.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

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6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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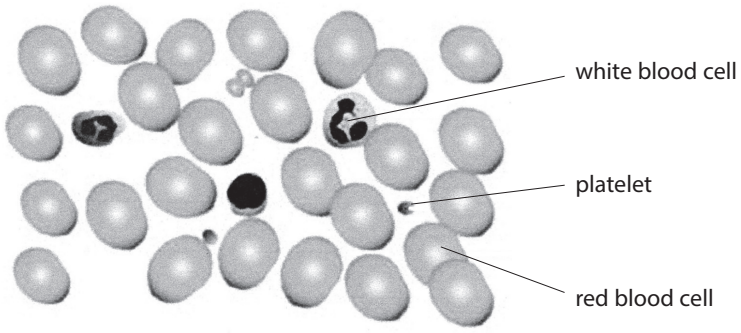
9. Question 6 · part b.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.



(i) Describe how the slide of human blood could be prepared.

(3)

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9. Question 6 · part b.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Discuss the accuracy of the student's diagram.

(4)

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(iii) Explain why the student would be advised to use a prepared slide rather than making one from their own blood.

(2)

(Total for Question 6 = 15 marks)

9. Question 6 · part b.iii · 2 marks · Working and answer

Working and answer

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10. Question 3 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

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2 .....

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3 .....

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(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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10. Question 3 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

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(d) Oral rehydration is a method used to treat the symptoms of Ebola.

Explain why this method is an effective treatment.

(3)

(e) Fruit bats are thought to be a natural reservoir for the Ebola pathogen.

Suggest what is meant by the term **reservoir**.

(2)

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(Total for Question 3 = 12 marks)

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10. Question 3 · part e · 2 marks · Working and answer

Working and answer

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11. Question 3 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

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(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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11. Question 3 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

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(d) Oral rehydration is a method used to treat the symptoms of Ebola.

Explain why this method is an effective treatment.

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Suggest what is meant by the term **reservoir**.

(2)

(Total for Question 3 = 12 marks)

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11. Question 3 · part c · 3 marks · Working and answer

Working and answer

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12. Question 4 · part a.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.i only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

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(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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12. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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13. Question 4 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

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(i) Give one other symptom of a pinworm infection.

(1)

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(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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13. Question 4 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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13. Question 4 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

(e) Suggest how an infected person and other people in the house can help prevent the spread of a pinworm infection (lines 13 to 14).

(2)

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(Total for Question 4 = 12 marks)

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Working and answer